Industrial Skill Framework (IndSF) for Semiconducter Technical and Engineering Staff by Job Area

Focus Area	Job Area	Job Position	Job Description	Soft Skills	Technical Skills
Technical	Manufacturing Operations	Operator	Operates machines and performs housekeeping in cleanroom,follows SOPs.	Communication, Teamwork, Discipline, Lifelong Learning	Machine Operation, 5S/Cleanroom, Safety (OSHA), Quality Basics, Good Manufacturing Practices
Technical	Manufacturing Operations	Senior Operator	Oversees production lines; trains junior operators; ensures compliance.	Communication, Problem Solving, Supervisory Skills	Equipment Operation, Maintenance Basics, Cleanroom/ESD Practices, Quality Control
Technical	Manufacturing Operations	Line Leader	Leads team of operators, ensures productivity, resolves process issues.	Leadership, Teamwork, Problem Solving, Communication	Production Coordination, Machine Troubleshooting, Quality Compliance, Reporting
Technical	Manufacturing Operations	Technical Assistant	Supports engineers; sets up and maintains equipment, assists in process monitoring.	Teamwork, Communication, Problem Solving	Equipment Calibration, Preventive Maintenance, SPC/Minitab, Data Collection
Technical	Manufacturing Operations	Technician	Performs maintenance, repair and equipment setup for production.	Time Management, Problem Solving, Communication	PLCs, Motors & Control Systems, Preventive Maintenance, Electrical/Pneumatic Systems
Technical	Manufacturing Operations	Senior Technician	Supervises maintenance team; leads troubleshooting; mentors junior techs.	Leadership, Analytical Thinking, Teamwork	Advanced Troubleshooting, Electrical/Mechanical Diagnostics, Quality Improvement
Technical	Manufacturing Operations	Manufacturing Trainer	Designs and delivers internal manufacturing training programmes.	Communication, Presentation, Coaching, Teamwork	Training Delivery, Lean Manufacturing, Quality Systems, Safety Compliance
Engineering	Factory Automation	Assistant Factory Automation Engineer	Supports design and maintenance of automation systems; assists with PLC and control system programming.	Problem Solving, Attention to Detail, Adaptability	PLC, Ladder Logic, C++, Python, Data Analysis, ESD Awareness
Engineering	Factory Automation	Factory Automation Engineer	Designs, implements, and maintains automated systems for process efficiency.	Analytical Thinking, Collaboration, Adaptability	Robotics, AI Tools (Power BI, UI Path), Data Analytics, Control Systems Integration
Engineering	Factory Automation	Senior Factory Automation Engineer	Leads automation strategy; integrates robotics, IoT and control systems; mentors team.	Leadership, Project Management, Influencing Skills	IoT, Embedded Systems, FPGA, AI Automation, Safety & Compliance (LOTO, HIRARC)
Engineering	Facilities	Assistant Facilities Engineer	Supports maintenance of utilities (HVAC, electrical, plumbing) and infrastructure systems.	Communication, Teamwork, Adaptability	Building Systems, HVAC, Plant Utilities Treatment (PUT), AutoCAD
Engineering	Facilities	Facilities Engineer	Oversees building systems and utilities; ensures operational reliability and compliance.	Supervisory Skills, Conflict Resolution, Analytical Thinking	HVAC, Safety Standards (OSHA, UBBL), CAD Design, Data Analysis
Engineering	Facilities	Senior Facilities Engineer	Leads large-scale facility upgrades, energy conservation and ESG initiatives.	Stakeholder Management, Leadership, Conflict Resolution	Energy Management, ESG Practices, Project & Vendor Management, Safety Compliance
Engineering	Equipment	Assistant Equipment Engineer	Assists in equipment installation, preventive maintenance and troubleshooting.	Team Collaboration, Adaptability, Time Management	Equipment Setup, Preventive Maintenance, Data Analytics, Safety Awareness
Engineering	Equipment	Equipment Engineer	Manages equipment performance and reliability; leads technical support to production.	Problem Solving, Communication, Analytical Skills	Equipment Calibration, DOE, Statistical Process Control, Supplier Management
Engineering	Equipment	Senior Equipment Engineer	Leads equipment improvement projects and advanced troubleshooting.	Leadership, Coaching, Project Management	Advanced Equipment Diagnostics, Robotics, Automation Integration, AI Tools
Engineering	Process	Assistant Process Engineer	Supports process setup and optimisation, data collection, yield tracking.	Communication, Analytical Thinking	Process Control, Data Analysis (SPC), DOE
Engineering	Process	Process Engineer	Develops and improves process parameters; ensures yield and quality consistency.	Problem Solving, Collaboration	SPC, FMEA, Process Mapping, Continuous Improvement
Engineering	Process	Senior Process Engineer	Leads process optimisation projects and cross-departmental improvement initiatives.	Leadership, Analytical Thinking	Advanced DOE, Six Sigma, AI-based Process Optimisation
Engineering	Quality	Assistant Quality Engineer	Supports quality inspection, control documentation, and audits.	Teamwork, Communication	ISO9001 Awareness, Quality Control Tools
Engineering	Quality	Quality Engineer	Manages quality systems, root-cause analysis, and customer audit responses.	Problem Solving, Analytical Thinking	APQP, PPAP, FMEA, SPC, MSA
Engineering	Quality	Senior Quality Engineer	Leads quality strategy, drives process reliability and compliance programs.	Leadership, Communication	Advanced Statistical Methods, Lean Six Sigma, Risk Management
Engineering	Test	Assistant Test Engineer	Conducts test setup and validation for semiconductor products.	Attention to Detail, Problem Solving	Test Equipment Setup, Data Analysis
Engineering	Test	Test Engineer	Develops test methods, analyses test data, improves efficiency.	Communication, Collaboration	Test Automation, Data Interpretation, Statistical Tools
Engineering	Test	Senior Test Engineer	Leads test development strategy, interfaces with design and product teams.	Leadership, Analytical Skills	Advanced Test Automation, Data Analytics, Quality Tools (FMEA, SPC)
Engineering	Product	Product Engineer	Supports production transfer, product qualification and design validation.	Project Management, Teamwork	Product Qualification, Data Correlation, Design Validation
Engineering	Product	Senior Product Engineer	Leads product development, integrates design and manufacturing improvements.	Leadership, Stakeholder Management	Product Lifecycle Management, DFM, FMEA, Reliability Testing
Engineering	NPI / Integration	Assistant NPI Engineer	Supports new product introduction builds (EVT/DVT); coordinates with process and quality teams.	Communication, Teamwork, Adaptability	NPI Process Flow, Product Qualification, Data Correlation, DOE
Engineering	NPI / Integration	NPI Engineer	Manages product ramp-up and ensures seamless transfer to production.	Project Management, Problem Solving	Design of Experiments (DOE), FMEA, Process Optimisation, Product Validation
Engineering	NPI / Integration	Senior NPI / Integration Engineer	Leads integration projects; drives manufacturability, yield and performance improvements.	Leadership, Stakeholder Management, Analytical Thinking	DFM/DFA, Reliability Testing, Failure Analysis, Product Lifecycle Management
Engineering	Package / Product Development (PDD)	Assistant Packaging Engineer	Supports package design and assembly process optimisation.	Analytical Thinking, Communication	Package Design, CAD Layout, Material Characterisation
Engineering	Package / Product Development (PDD)	Packaging Engineer	Designs new semiconductor package types; coordinates cross-functional teams.	Problem Solving, Collaboration	Package Modelling (ANSYS, SolidWorks), Thermal Analysis, FEA, DOE
Engineering	Package / Product Development (PDD)	Senior Packaging Engineer	Leads advanced packaging and 3D/SiP innovations.	Leadership, Project Management	Advanced Packaging, Flip Chip, TSV, Process Simulation, Reliability Validation
Engineering	Test Development & Integration (TDI)	Assistant TDI Engineer	Supports test hardware/software development and validation for new products.	Teamwork, Analytical Thinking	Test Program Debugging, Test Fixture Design, LabView/Python Scripting
Engineering	Test Development & Integration (TDI)	TDI Engineer	Develops test solutions for IC validation; integrates ATE systems with product specs.	Communication, Problem Solving	ATE Systems (Teradyne, Advantest), Test Data Analytics, Correlation Studies
Engineering	Test Development & Integration (TDI)	Senior TDI Engineer	Leads cross-functional test integration; drives yield improvement and failure analysis.	Leadership, Collaboration, Analytical Thinking	Test Methodology Design, Data Automation, Yield Analysis, Reliability Testing